

Python Practice Problems

Exercise 2 — Fundamentals Continued

Topics covered: Strings · Numbers & Math · User Input · Lists & List Functions · Loops & Exit Conditions

20 Problems · 6 Easy · 14 Medium

Part A — Easy Problems

- EASY** **Greeting by Time of Day**
Ask the user to enter the current hour (0–23). Print **Good Morning** if it is before 12, **Good Afternoon** if it is before 18, and **Good Evening** otherwise.
- EASY** **Initials Generator**
Ask the user to enter their full name (e.g. John Michael Doe). Print their initials separated by dots (e.g. J.M.D). Use `split()` to break the name into parts.
- EASY** **Swap Two Numbers**
Ask the user for two numbers. Without using a third variable, swap their values and print both numbers before and after the swap.
- EASY** **List from Input**
Ask the user to enter 5 numbers, one at a time. Store them in a list. Print the list, then print the same list in reverse order using slicing.
- EASY** **String Repeat Counter**
Ask the user to enter a word. Ask them to enter a letter. Count how many times that letter appears in the word using the `count()` string method and print the result.
- EASY** **Absolute Difference**
Ask the user for two numbers. Print the absolute difference between them (always a positive value) using `abs()`.

Part B — Medium Problems

- MEDIUM** **Number Guessing Game**
Pick a secret number (hardcode it as a variable). Ask the user to keep guessing until they get it right, using a `while True` loop with a `break`. After each wrong guess, print whether the secret number is **higher** or **lower**. Print the number of attempts taken when they win.
- MEDIUM** **Running Total**
Ask the user to keep entering numbers in a loop. After each entry, print the running

total so far. The loop should exit when the user types **done**. Print the final total and count of numbers entered.

9. **MEDIUM** Password Validator

Ask the user to enter a password in a loop. Accept the password only if it meets all of these rules:

- At least 8 characters long
- Contains at least one digit
- Contains at least one uppercase letter

Print which rule(s) failed on each attempt. Exit the loop once a valid password is entered.

10. **MEDIUM** FizzBuzz with Input

Ask the user for a number **n**. Print every number from 1 to **n**. Replace multiples of 3 with **Fizz**, multiples of 5 with **Buzz**, and multiples of both with **FizzBuzz**.

11. **MEDIUM** List Duplicate Remover

Ask the user to enter 10 words, one at a time, and store them in a list. Print the original list. Then remove all duplicates while preserving the original order and print the cleaned list. *Hint: use a second empty list and check with `in`.*

12. **MEDIUM** Sentence Statistics

Ask the user to type a sentence. Print:

- Total number of characters (including spaces)
- Total number of words
- Total number of sentences (count full stops, exclamation marks, and question marks)
- The longest word in the sentence

13. **MEDIUM** Math Quiz

Generate 5 random addition questions using numbers between 1 and 20. Show each question to the user and ask for their answer in a loop. Keep track of how many they got right and print the final score at the end (e.g. You scored 4 out of 5). *Hint: use `import random` and `random.randint()`.*

14. **MEDIUM** Inventory Manager

Start with a pre-filled list of 5 items (e.g. ["apple", "bread", "milk", "eggs", "butter"]). Run a loop that gives the user a menu:

- a) Add item to end of list
- b) Insert item at a specific position
- c) Remove item by name

d) View sorted inventory

e) Quit

Use `append()`, `insert()`, `remove()`, and `sort()`.

15. MEDIUM Hypotenuse Calculator

Ask the user for the two shorter sides (**a** and **b**) of a right-angled triangle. Using `math.sqrt()` and `math.pow()`, calculate the hypotenuse (**c**) using Pythagoras' theorem: $c = \sqrt{a^2 + b^2}$. Print the result rounded to 4 decimal places.

16. MEDIUM Contact Book

Use two parallel lists: one for names and one for phone numbers. Run a loop with a menu:

a) Add a contact (name + number)

b) Search for a contact by name

c) Delete a contact by name

d) View all contacts

e) Quit

When searching, print the name and number. Use `index()` to find and remove entries.

17. MEDIUM String Encoder

Ask the user to enter a message. Encode it by replacing every vowel with the next vowel in the sequence (**a**→**e**, **e**→**i**, **i**→**o**, **o**→**u**, **u**→**a**). Preserve case. Print the encoded message. Then also print the decoded version (reverse the substitution).

18. MEDIUM Number to Words

Ask the user to enter a number between 1 and 19. Print the English word for it (e.g. 7 → **seven**). Store the word equivalents in a list where the index matches the number. Run this in a loop until the user types **quit**.

19. MEDIUM Leaderboard

Ask the user to enter 6 player names and their scores, one pair at a time. Store names in one list and scores in another. Then print the leaderboard sorted from highest to lowest score, with a rank number next to each entry (e.g. **#1 Alice -- 980**). *Hint: sort the scores list with `sort(reverse=True)` and reorder names accordingly.*

20. MEDIUM Loan Repayment Counter

Ask the user for a loan amount and a fixed monthly repayment amount. Using a loop, subtract the repayment each month and print the remaining balance after each payment. Stop the loop when the balance reaches zero or below. Print how many months it took to repay the loan in full. Use `math.ceil()` for any partial final payment.

Tip: For all loop problems, practice using `while True` with a `break` as your exit condition — it is the clearest and most Pythonic pattern for input-driven loops.